

Column vectors



1

a $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$

b $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$

2

a $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

b $\begin{bmatrix} 3 \\ -3 \end{bmatrix}$

c $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

3

a i $\begin{bmatrix} 6 \\ 4 \end{bmatrix}$

ii $\begin{bmatrix} 6 \\ -4 \end{bmatrix}$

b i $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$

ii $\begin{bmatrix} -5 \\ -3 \end{bmatrix}$

4

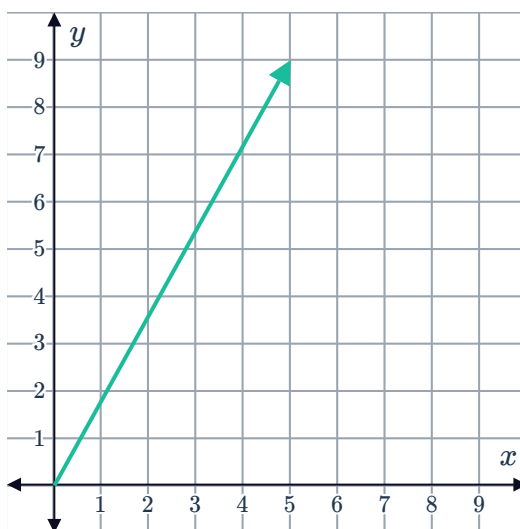
a $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$

b $\begin{bmatrix} -7 \\ 3 \end{bmatrix}$

c $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$

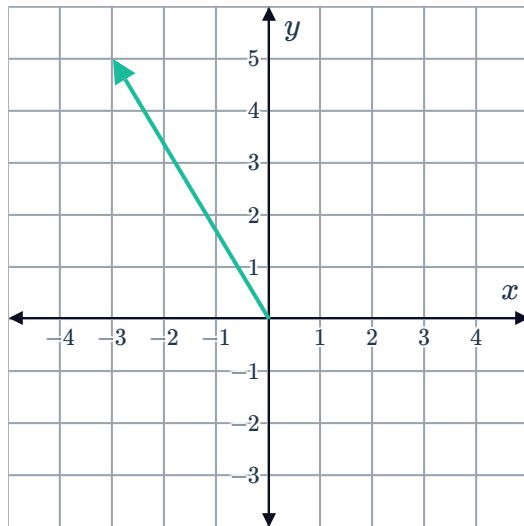
d $\begin{bmatrix} 10 \\ 1 \end{bmatrix}$

5

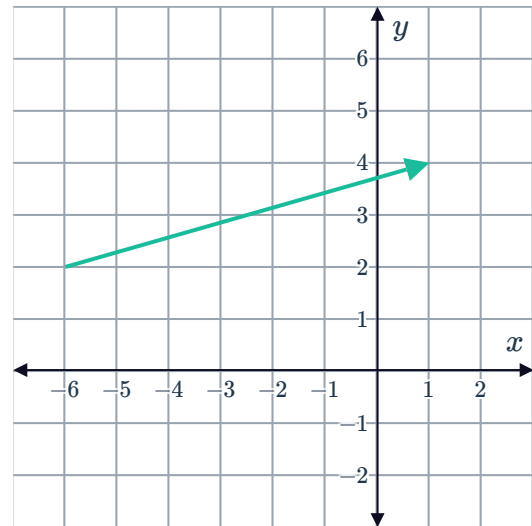


6

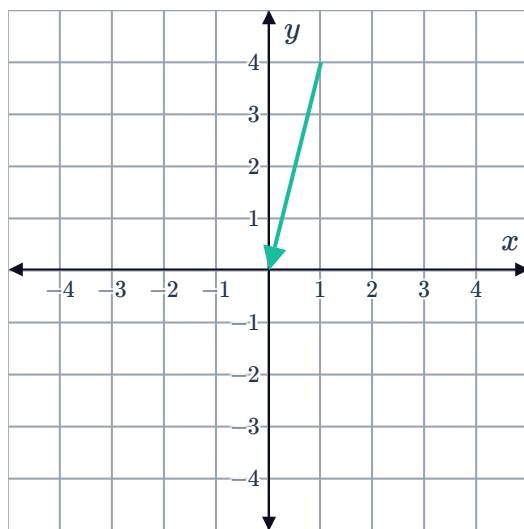
a



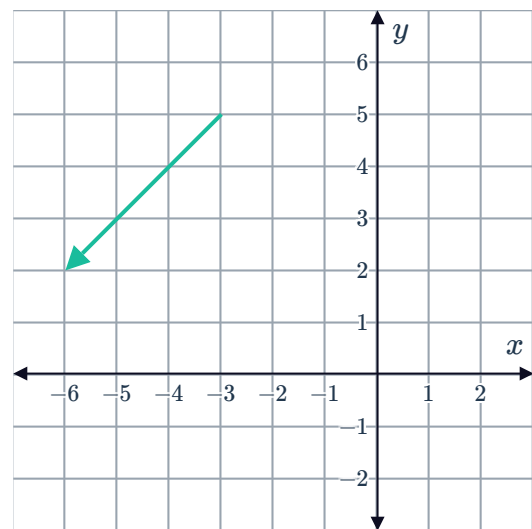
b



c



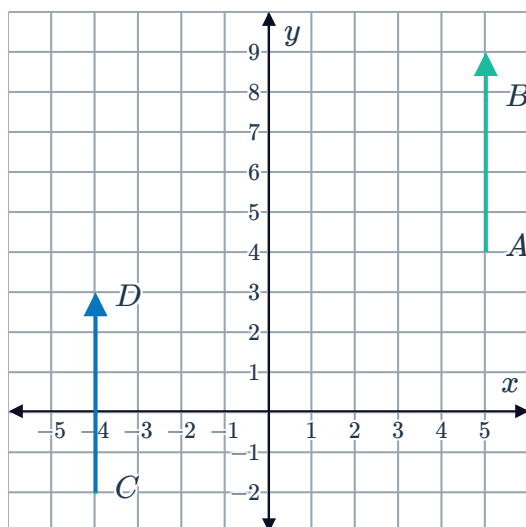
d



7

a

i

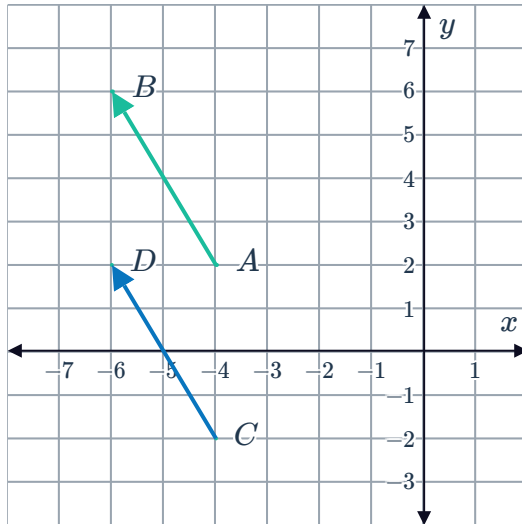


ii Yes

b i

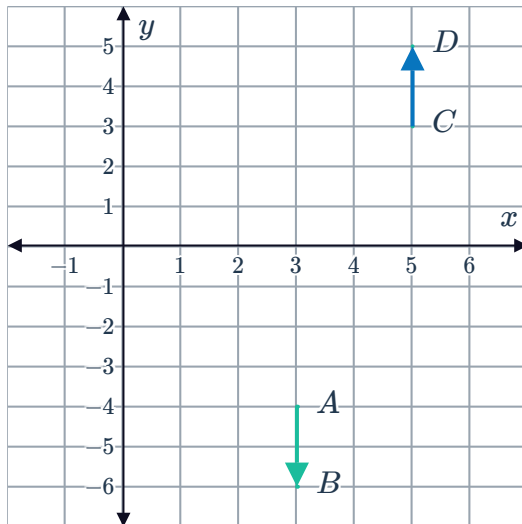


ii Yes



c i

ii No



8

a Yes

b Yes

c No

d No

9 800 km north

Magnitude of a vector (Extended)

10 5 units

11 5 units

12 6 units

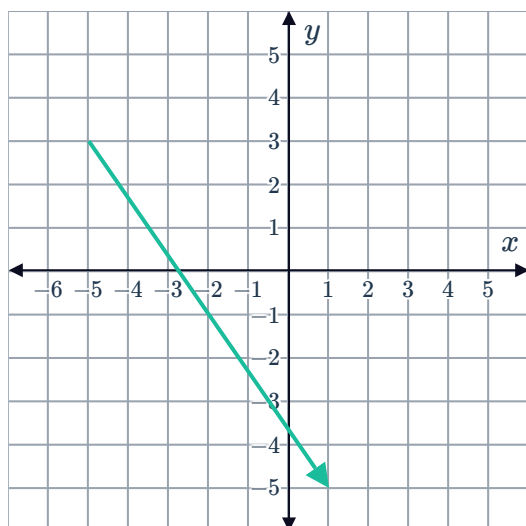
13



- a 15 units
- b 51 units
- c No, because the displacement is just the direct distance from the beginning point to the end point of the path.

14

a



b 10 units

15 True

16

- a 7 units
- b $\sqrt{74}$ units
- c 17 units
- d 17 units
- e 6 units
- f 6 units

17 $4\sqrt{5}$ units

18 13 units

19 $x = 3, -3$

20 20 km

21 13km